MAIS PAC

IPhO 02.06: Electrostatics and Steady Currents · Lecture + Problems

Electrostatics and Steady Currents

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. Charge, field and potential
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1. Charge, field and potential

Charge is quantised, conserved, and comes in two signs. The force between two point charges is Coulomb's law:

$$F = (1)/(4\pi\varepsilon_0)(q_1q_2)/(r^2),\ (1)/(4\pi\varepsilon_0) = 8.99\times10^{9}\ \{N,m^2,C^{-2}\}. \quad (1)$$

It is mathematically the same law as gravitation, except that charge can be negative and the force can be screened (in a conductor, or through a dielectric).

Rather than tracking forces, physicists use **fields**. The electric field is

$$E = (F)/(q), \quad (2)$$

and the potential is the potential energy per unit charge,

$$V = (U)/(q), E = - V. \tag{3}$$

The field is a vector and the potential is a scalar, so the potential is usually easier to add: $V = \Sigma kq_i/r_i$, while E must be added as vectors.

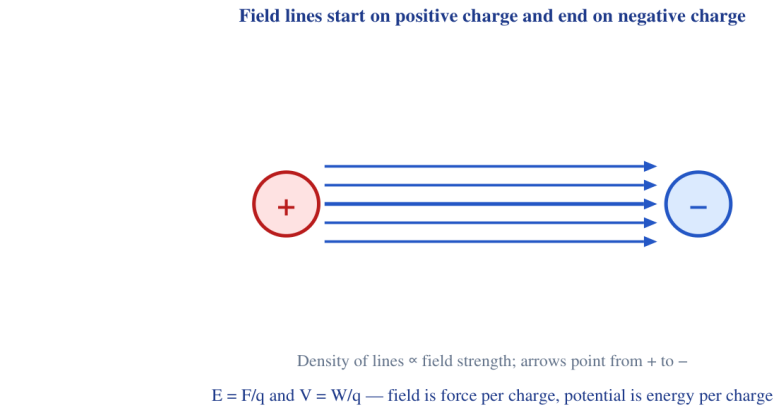


Figure 1. The field is a vector at every point; the potential is a scalar. The field points along the downhill direction of the potential.

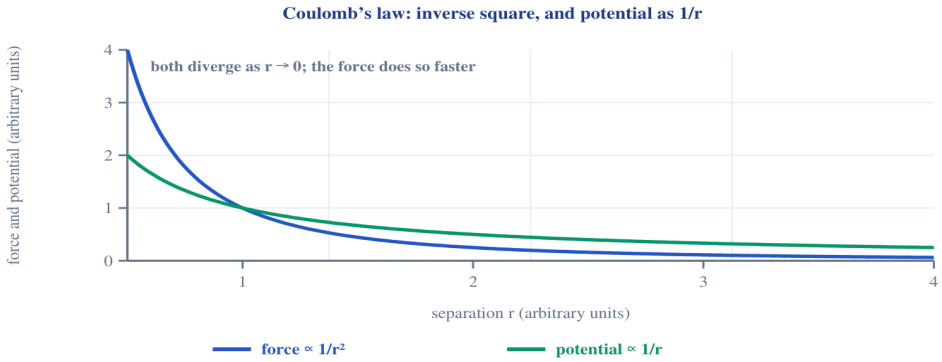


Figure. Force falls as $1/r^2$ while potential energy falls only as 1/r — that difference is why bound states exist.

2. Gauss's law and the method of symmetry

The flux of the field through a closed surface equals the enclosed charge divided by ε_0:

$$E\cdot d A = (Q_({enc}))/(\varepsilon_0). \tag{4}$$

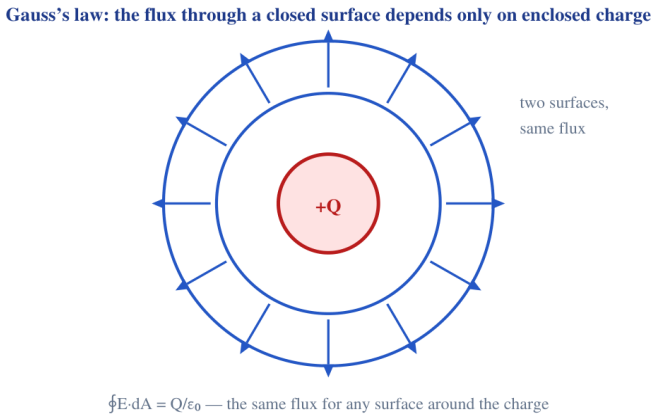


Figure 2. Gauss's law is only useful when symmetry makes E constant over the surface: spheres, cylinders, infinite planes.

Gauss's law is always true and almost never useful **directly**; it becomes a calculation only when the symmetry makes E constant over the surface. The three classic results:

- Spherical shell of charge Q: E = 0 inside, E = kQ/r² outside — exactly as if all charge sat at the centre.
- Uniformly charged solid sphere: E = kQr/R³ inside, growing linearly from the centre.
- Infinite plane with surface charge density σ: E = σ/(2ε_0), uniform.

Conductors in equilibrium have E=0 inside, so all excess charge sits on the surface, and the surface is an equipotential. Sharp points have larger surface charge density and therefore stronger fields — the reason lightning rods work.

3. Capacitance and stored energy

A capacitor stores charge proportional to the applied voltage, Q = CV, and holds energy

$$U = (1)/(2)\ CV^2 = (Q^2)/(2C). \tag{5}$$

For a parallel-plate capacitor of area A and separation d,

$$C = (\varepsilon_0\ A)/(d)\ (\{\text{multiplied by }\}\ \varepsilon_r\ \{\ \text{with a dielectric}\}). \tag{6}$$

Capacitors in parallel add, C = C_1 + C_2; in series they combine reciprocally, (1)/(C) = (1)/(C_1)+(1)/(C_2) — the opposite of resistors, because what adds is charge, not voltage drop.

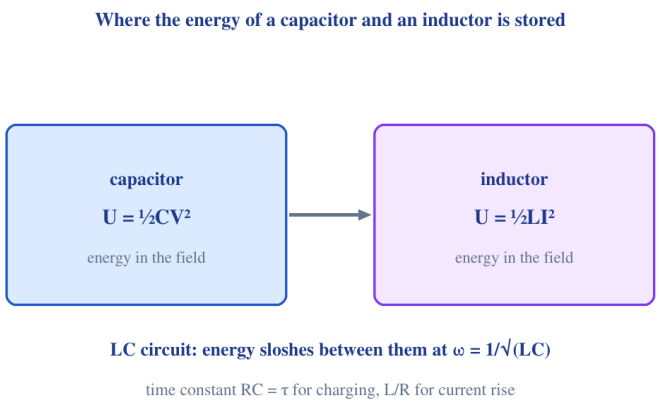


Figure 5. A capacitor is the electrical analogue of a spring, an inductor the analogue of mass: the oscillation frequency is 1/√(LC) exactly as 1/√(m/k) in mechanics.

4. Current, resistance and Ohm's law

Current is charge per unit time, I = dQ/dt. For an ohmic conductor,

$$V = IR,\ R = (\rho\ L)/(A), \tag{7}$$

where ρ is the resistivity. Power dissipated is

$$P = IV = I^2R = (V^2)/(R). \tag{8}$$

A non-ohmic element has a curved I–V characteristic (a filament lamp, a diode); then R = V/I still defines a resistance at each point, but it is not a constant.

5. Kirchhoff's rules and circuit analysis

Two laws, both conservation statements:

- **Junction rule:** Σ I_({in}) = Σ I_({out}) — charge conservation.
- **Loop rule:** Σ Δ V = 0 around any closed loop — energy conservation.

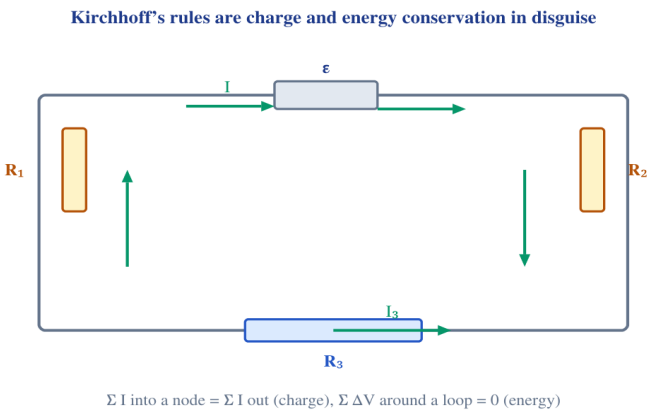


Figure 4. Choose a current direction for each branch, apply the junction rule, then the loop rule. A negative answer simply means the current flows the other way.

Working method: label a current direction in every branch (the sign is irrelevant as long as you are consistent), write one junction equation per independent node, then one loop equation per independent loop. A circuit with n nodes and b branches needs b independent equations.

Two rewriting tools solve most problems without algebra:

- **Series/parallel reduction** when the network allows it.
- **The potential divider**: with no load, V_({out}) = V(R_2)/(R_1+R_2). With a load, replace R_2 by R_2 R_L — the ideal formula fails badly once the load draws comparable current.

Interactive experiment: The loaded potential divider — open the online handout to use it.

Real sources have internal resistance r, so V_({terminal}) = ε - Ir; maximum power transfer to a load happens at R_L = r, where the efficiency is only 50%.

The electrostatics checklist. Symmetry first (then Gauss) → potential rather than field when adding contributions → energy U = (Q²)/(2C) or (1)/(2)CV² for capacitors → junction and loop rules for circuits → check limits (distance to zero, load to infinity, dielectric to vacuum).

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Three charges in a line. Charges +2.0 μ{C} and +3.0 μ{C} are 1.2 {m} apart. (a) Find the force between them. (b) Where must a third charge be placed on the line so that it feels no net force? (c) Find the electric potential at that point.

Worked solution →

[1] Problem 2 — Field of a charged sphere. A solid insulating sphere of radius R carries a total charge Q uniformly distributed. (a) Find E(r) inside and outside. (b) Find V(r) relative to infinity, both inside and outside. (c) Sketch both and explain why E is discontinuous in slope at the surface while V is continuous.

Worked solution →

[2] Problem 3 — Capacitors in series and parallel. Three capacitors C_1 = 2.0 μ{F}, C_2 = 3.0 μ{F} and C_3 = 6.0 μ{F} are connected to a 12 {V} supply. (a) Find the equivalent capacitance if they are all in parallel, and the total energy stored. (b) Find the equivalent capacitance in series, and the total energy. (c) Explain why the stored energy differs so much for the same supply voltage.

Worked solution →

[2] Problem 4 — Charging a capacitor. A capacitor C is charged through a resistor R from a battery of emf ε. (a) Show that q(t) = Cε(1-e^(-t/RC)). (b) Show that the energy supplied by the battery is Cε², while the energy stored is only half of that, and find where the other half goes. (c) Find the fraction of the final charge reached after one time constant.

Worked solution →

[2] Problem 5 — A Wheatstone bridge. Four resistors form a bridge across a 6.0 {V} supply: R_1 = 10 Ω and R_2 = 20 Ω in the left arm, R_3 = 15 Ω and R_4 = 30 Ω in the right arm, with a galvanometer between the midpoints. (a) Show that the bridge is balanced, i.e. no current flows through the galvanometer. (b) Derive the general balance condition. (c) Change R_4 to 33 Ω: find the direction of the current through the galvanometer and explain the qualitative reasoning that avoids solving the full network.

Worked solution →

[2] Problem 6 — Drift, mean free time and the electron gas. Copper has 8.5×10^(28) free electrons per cubic metre and resistivity 1.7×10^(-8) Ω,{m}. (a) Find the drift speed in a 1.0 {mm²} wire carrying 2.0 {A}. (b) Estimate the mean time between collisions using the Drude model. (c) Explain why the drift speed is so much smaller than the electrons' random speeds, and why this is not a paradox.

Worked solution →

[2] Problem 7 — Two spheres connected by a wire. Two isolated conducting spheres of radii r_1 and r_2 are far apart and carry charges Q_1 and Q_2. They are then connected by a thin wire. (a) Find the final charges. (b) Show that the surface charge densities and the fields at the two surfaces are in the ratio r_2:r_1 and (r_2/r_1)². (c) Relate the result to the action of a lightning rod.

Worked solution →

[2] Problem 8 — Internal resistance and maximum power. A battery has emf 12 {V} and internal resistance 0.50 Ω. (a) Find the terminal voltage for a load of 5.0 Ω. (b) Find the load that extracts maximum power and the power then. (c) Show that at maximum power the battery dissipates exactly as much as the load, and explain why a real power supply is not designed this way.

Worked solution →

[2] Problem 9 — Potentiometer. A potentiometer consists of a uniform wire of length 1.00 {m} and resistance 12 Ω connected across a 2.0 {V} accumulator, with a cell of unknown emf in the second circuit. The balance point is at 62.0 {cm}. (a) Find the unknown emf. (b) Explain precisely why the method does **not** require knowing the resistance of the wire, only its uniformity. (c) Find the change in the balance length if the accumulator's emf drops to 1.9 {V}.

Worked solution →

[3] Problem 10 — A sliding contact and a nonlinear element. A rod of resistance R_0 slides on rails in a circuit so that its resistance in the circuit varies linearly with position, R(x) = R_0(1+3x/L), and is connected in series with a diode whose forward voltage is a constant 0.70 {V} once it conducts. The supply is a 9.0 {V} battery with negligible internal resistance. (a) Find the current as a function of x if the diode is forward-biased. (b) Find the value of x at which the diode stops conducting, given R_0 = 6.0 Ω. (c) Explain why Kirchhoff's loop rule still applies to a nonlinear element and why the voltage–current characteristic is the only extra information needed.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

MIT OpenCourseWare 8.02, Electricity and Magnetism

Full lecture notes and problem sets for fields, circuits and induction.